

# The cross-base identity for OEIS A125409, proved

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## Abstract

OEIS A125409 carries an empirical identity relating its counts to those of the entry one base smaller, contributed by R. H. Hardin. It is true, it is true in the exact range the entry names, and that range is sharp. The proof is four steps and needs no computation: the difference between consecutive bases counts the admissible cyclic sequences that USE the new largest digit, every admissible cyclic sequence has range at most  $d\lfloor n/2 \rfloor$ , so once the base is large enough those sequences are a translate of the admissible cyclic sequences over  $\mathbf{Z}$  with maximum 0, and those are in bijection with the step sequences summing to zero.

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## 1 The conjecture

OEIS A125409 is “Number of base 21 circular  $n$ -digit numbers with adjacent digits differing by 6 or less.”. It has offset 0 and begins

1, 21, 231, 2121, 22547, 244601, 2714271, 30517333, ...

The entry states, as an empirical claim and never marked settled:

[Empirical]  $a(\text{base}, n) = a(\text{base}-1, n) + F(6)$  for  $\text{base} \geq 6\lfloor n/2 \rfloor + 1$  and  $F(d)$  is the largest coefficient in  $(1+x+\dots+x^{2d})^n$

As of the “Last modified” line on the live entry (Aug 12 2026 19:06:33, revision 14) this is still recorded as empirical, and nothing on the entry records it as proved.

Read as the entry writes it,  $a(\text{base}, n)$  is the count this family assigns to the stated base, “ $\text{int}(n/2)$ ” is  $\lfloor n/2 \rfloor$ , and  $F(d)$  is defined on the entry itself as the largest coefficient of  $(1+x+\dots+x^{2d})^n$ . For this entry the base is 21 and the permitted difference is  $d=6$ , so the claim is that

$$a_{21}(n) - a_{20}(n) = F_6(n) \quad \text{for every } n \text{ with } 6\lfloor n/2 \rfloor + 1 \leq 21,$$

that is, for  $1 \leq n \leq 7$ .

## 2 What the entry counts

A base- $b$  circular  $n$ -digit number, in this family’s sense, is a tuple  $(d_1, \dots, d_n) \in \{0, 1, \dots, b-1\}^n$  with  $|d_i - d_{i+1}| \leq d$  for every  $i$ , indices read modulo  $n$  so that the pair  $(d_n, d_1)$  is included. Leading zeros are counted: requiring  $d_1 \neq 0$  gives 4, 11, 25 where A124698 gives 5, 13, 29. Write  $T_d(b, n)$  for the number of them. Equivalently  $T_d(b, n) = \text{tr } M^n$  with  $M$  the  $b \times b$  matrix that has  $M_{uv} = 1$  exactly when  $|u - v| \leq d$ , since a cyclic tuple is a closed walk in that graph together with a choice of where it starts. Evaluated exactly in integer arithmetic,  $T_6$  at base

21 reproduces all 18 terms A125409 publishes, and at base 20 all the terms the entry one base smaller publishes.

At  $n = 0$  these entries write  $a(0) = 1$  while the count of empty tuples read as a trace is  $b$ . That is a convention at a single index; the identity below is stated and proved for  $n \geq 1$ , which is the range in which the entry's own data lives beyond its first term.

### 3 The identity

**Theorem 1.** *For every  $d \geq 1$ , every  $n \geq 1$  and every  $b \geq d\lfloor n/2 \rfloor + 1$ ,*

$$T_d(b, n) - T_d(b - 1, n) = F_d(n),$$

where  $F_d(n) = [x^{dn}](1 + x + \dots + x^{2d})^n$  is the largest coefficient of that polynomial. In particular  $a_{21}(n) - a_{20}(n) = F_6(n)$  for  $1 \leq n \leq 7$ , which is the entry's claim.

*Proof. Step 1.* An admissible tuple over  $\{0, \dots, b-1\}$  either uses the value  $b-1$  or does not, and those that do not are exactly the admissible tuples over  $\{0, \dots, b-2\}$ . Hence  $T_d(b, n) - T_d(b-1, n)$  is the number of admissible tuples over  $\{0, \dots, b-1\}$  in which  $b-1$  occurs.

*Step 2.* Every admissible tuple satisfies  $\max_i d_i - \min_i d_i \leq d\lfloor n/2 \rfloor$ . Fix a position  $i$  where the maximum is attained and a position  $j$  where the minimum is. The two arcs of the cycle joining  $i$  to  $j$  have lengths summing to  $n$ , so one of them has length  $L \leq \lfloor n/2 \rfloor$ ; along an arc of length  $L$  each step changes the value by at most  $d$ , so the two values differ by at most  $dL \leq d\lfloor n/2 \rfloor$ .

*Step 3.* A tuple counted in Step 1 has maximum exactly  $b-1$ , so by Step 2 all of its values lie in  $[b-1-d\lfloor n/2 \rfloor, b-1]$ . The hypothesis  $b \geq d\lfloor n/2 \rfloor + 1$  says precisely that this window lies inside  $\{0, \dots, b-1\}$ , so no constraint of the base is felt: subtracting  $b-1$  from every entry is a bijection between the tuples counted in Step 1 and the tuples  $(c_1, \dots, c_n) \in \mathbf{Z}^n$  with  $|c_i - c_{i+1}| \leq d$  cyclically and  $\max_i c_i = 0$ . That set does not depend on  $b$ .

*Step 4.* Translation acts freely on the admissible tuples over  $\mathbf{Z}$ , and each orbit contains exactly one tuple with maximum 0 and exactly one with  $c_1 = 0$ . So the tuples of Step 3 are in bijection with the admissible tuples having  $c_1 = 0$ , which are determined by their step vector  $(s_1, \dots, s_n)$ ,  $s_i = c_{i+1} - c_i$  read cyclically: any  $(s_1, \dots, s_n) \in \{-d, \dots, d\}^n$  with  $s_1 + \dots + s_n = 0$  arises from exactly one such tuple, and conversely. Their number is the constant term of  $(x^{-d} + \dots + x^d)^n$ , that is  $[x^{dn}](1 + x + \dots + x^{2d})^n = F_d(n)$ , which is the largest coefficient of that polynomial because the coefficients are symmetric about  $x^{dn}$  and unimodal.  $\square$

**Remark 2.** The threshold is sharp. At  $b = d\lfloor n/2 \rfloor$  the window of Step 3 no longer fits inside the alphabet, the tuples of maximum  $b-1$  are cut off from below, and the difference falls short of  $F_d(n)$ ; this was checked directly for  $d = 1, 2, 3$  and every  $n \leq 8$ , and the first failing base is in each case exactly one below the entry's own threshold.

### 4 Verification

Three checks, each independent of the proof above.

First, the model reproduces all 18 terms A125409 publishes, exactly, in integer arithmetic; the reading of the entry's English is what is being tested there, since a misreading gives different counts.

Second, the identity was evaluated directly for every  $n$  in the entry's claimed range,  $1 \leq n \leq 7$ , with both sides computed from the definition and not from the theorem:

| $n$ | $T_6(21, n)$ | $T_6(20, n)$ | $F_6(n)$ |
|-----|--------------|--------------|----------|
| 1   | 21           | 20           | 1        |
| 2   | 231          | 218          | 13       |
| 3   | 2121         | 1994         | 127      |
| 4   | 22547        | 21078        | 1469     |
| 5   | 244601       | 227450       | 17151    |
| 6   | 2714271      | 2509508      | 204763   |
| 7   | 30517333     | 28044008     | 2473325  |

and the difference of the second and third columns is the fourth at every row shown and at every row up to  $n = 7$ .

Third, the theorem was tested outside the entry's range as well: the identity holds for every  $b \geq d\lfloor n/2 \rfloor + 1$  tried and fails immediately below, which is the sharpness recorded above.

## References

1. The OEIS Foundation Inc., *The On-Line Encyclopedia of Integer Sequences*, <https://oeis.org/A125409>.